INTRODUCTION

This open-ended laboratory experiment involves the design of a Frequency Modulator utilizing the Voltage controlled (VCO) section of the XR2206 integrated circuit. The XR2206 generates a frequency of oscillation that is directly proportional to the total timing current drawn from either Pin 7 or Pin 8.

PROBLEM ANALYSIS

The primary objective is to configure the XR2206 to operate at a designated centre frequency and to analyze the circuit's response when a sweep voltage is applied to modulate the frequency. This require mathematically deriving the unknow circuit Resistor values & subsequently calculating are modulated

signal parameters to evaluate the system's performance.

3. DESIGN REQUIREMENTS

- carrier Frequency (f_0): 45kHz
- Timing capacitor (C): 10nF
- Determine the values for the unknown circuit Resistor (R)

$$\Rightarrow f_0 = \frac{1}{RC} \text{ Hz}$$

$$\Rightarrow R = \frac{1}{(45\times10^3)(10\times10^{-9})}$$

$$\boxed{R = 2.2\text{k}\Omega}$$

- Measure the FM modulated signal parameters: f_{min}, f_{max}, frequency Deviation (Δf), modulation index (β).

Scanned with CamScanner

4- POSSIBLE SOLUTIONS

The base frequency of operation can be established by connecting an external timing capacitor C across pin 5 & 6, along with timing Resistor R connected to Pin 7 or 8. Frequency modulation is then activated / achieved by applying a control voltage (Vc) to the activated timing pin, which is linearly varies the timing current and, consequently, the output frequency.

5- PRELIMINARY DESIGN

The mathematical calculations for the component values are based on the standard operation formulas for the XR2206:

- The centre frequency is defined by the equation $f_0 = \frac{1}{RC}$
- Substituting the known parameters yields:

$$45 \times 10^3 = \frac{1}{R(10 \times 10^{-9})}$$

- Solving for the timing resistor gives

$$R = 2.2 k\Omega$$

- The frequency is also dependend on total timing current (I_T), given by

$$f = \frac{320\ I_T (mA)}{C(\mu F)}$$

- Solving for the timing circuit yields

$$I_T = \frac{45 \times (10.01)}{320} = 1.4 mA$$

- The internal voltage at the timing (Pin 7 or 8) is biased a +3V with respect to Pin 12. Assuming the 1.4mA current splits equally ($I = 0.7 mA$), the required Resistance R_t is calculated using ohm's law.

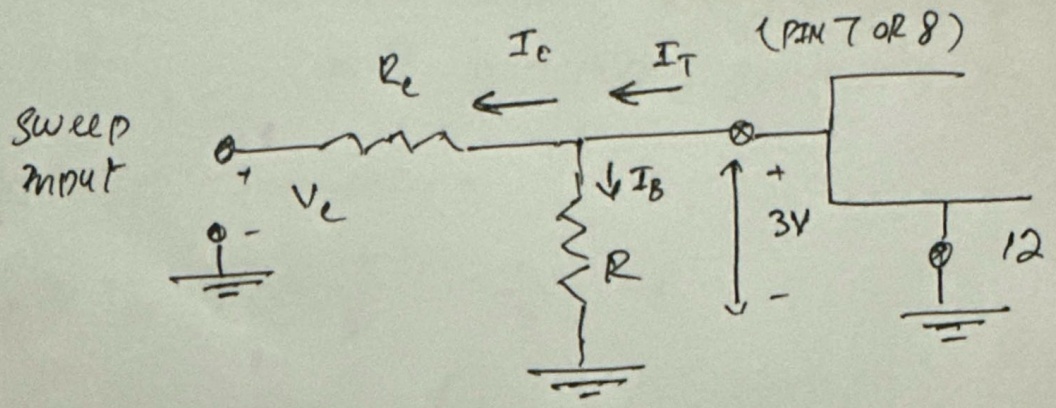
Scanned with CamScanner

$(V = IR)$:

$$R_c = \frac{3}{0.7} = 4.28\,k\Omega$$

The design establishes the condition that

$12 = R_c$



6. DESIGN DESCRIPTION

The circuit utilizes a frequency sweep configuration where the modulating sweep voltage V_c is applied across the Resistor R_c, while the Internal +3V bias drives the resistor R tied to Ground. The resulting operational frequency is related to applied control voltage V_c by the expression:

$$f = \frac{1}{RC}\left(1 + \frac{R}{Rc}\left(1 - \frac{V_c}{3}\right)\right).$$

Scanned with CamScanner

7- SOFTWARE SIMULATION:

Scanned with CamScanner

8- EXPERIMENTAL RESULTS:

The maximum & minimum frequencies were calculated for two distinct modulating signals:

- Case 1: At a control voltage of 2 V_{p-p} and modulating frequency 2kHz

Maximum Frequency:

$$f_{max} = \frac{1}{RC}\left(1 + \frac{R}{R_c}\left(1 - \frac{V_c}{3}\right)\right) Hz$$

$$= \frac{1}{(4.28\times10^{3})(10\times10^{-9})}\left[1 + 1\left(1 - \frac{(-1)}{3}\right)\right]$$

$$f_{max} = 62.517\ kHz$$

Minimum Frequency:

$$f_{min} = \frac{1}{RC}\left[1 + \frac{R}{R_c}\left(1 - \frac{V_c}{3}\right)\right]$$

$$= \frac{1}{(4.28\times10^{3})(10\times10^{-9})}\left[1 + 1\left(1 - \frac{1}{3}\right)\right]$$

$$f_{min} = 31.94\ kHz$$

Case 2 : At a control voltage of 4Vpp and modulating frequency of 4KHz

- Maximum frequency (f_{max}):

$$f_{max} = \frac{1}{RC}\left[1 + \frac{R}{R_c}\left(1 - \frac{V_c}{3}\right)\right]$$

$$f_{max} = \frac{1}{(4.28\times10^3)(10\times10^{-9})}\left[1 + 1\left(1 - \frac{(-4)}{3}\right)\right]$$

$$\boxed{f_{max} = 62.30\ KHz}$$

- Minimum Frequency (f_{min}):

$$f_{min} = \frac{1}{RC}\left[1 + \frac{R}{R_c}\left(1 - \frac{V_c}{3}\right)\right]$$

$$= \frac{1}{(4.28\times10^3)(10\times10^{-9})}\left[1 + 1\left(1 - \frac{4}{3}\right)\right]$$

$$\boxed{f_{min} = 31.15\ KHz}$$

9- PERFORMANCE ANALYSIS

Using the experimental results, the frequency deviation ($\Delta f = \frac{f_{max} - f_{min}}{2}$) and the modulation index $\beta = \frac{\Delta f}{f_m}$ were determined:

- For case 1 (2Vp-p , 2kHz):
  - $\Delta f = \frac{54.57 - 38.94}{2} = 7.815 kHz$
  - $\beta = \frac{7.815 kHz}{2 kHz} = 3.9075$
- For case 2 (4Vp-p , 4kHz):
  - $\Delta f = \frac{62.30 - 31.15}{2} = 15.57 kHz$
  - $\beta = \frac{15.57}{4kHz} \Rightarrow 3.8975$

Scanned with CamScanner

10- CONCLUSION:

The frequency modulator was successfully designed for a 45kHz carrier using the XR 2206 integerated circuit. The calculation demonstrates a proportional relationship between the applied control voltage and the resulting frequency deviation (from 2Vp-p at 2kHz to 4Vp-p at 4KHz), the modulation Index β remain nearly constant at approximately 3.9.

REFERENCES:

1- Lab Manual of 'Analogue and Digital communication'

Experiment 6.

2. XR2206 Monolithic Function Generator Datasheet, Exar corporation.

Scanned with CamScanner